

Art & Design

Year 3

Cambridge Early Years

Student Manual



Art & Design

Year 3 · Student Manual
The Bright Studio

A Cambridge Primary student book for Lower Primary. Written for pupils of about seven years old who are ready to look harder, make longer, and talk about what they have made.

Four tables in one bright studio: drawing, mosaic, painting and structure. Rufus the fox and Bruno the bear work beside you.

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Art & Design, Year 3 · Student Manual
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Interior 8.5 × 11 in on white stock. British English throughout. Printed so that pencil, crayon and glue can work on the page.

This book is meant to be used. Draw in it. Stick tiles in it. Get a little paint on it.

Safety: activities use washable paint, blunt tools and card. Cutting and any clay work need an adult nearby. Teachers follow their school safety policy.

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Welcome to the Bright Studio

There is a bright room with one long table.

On the table you will find pencils, soft charcoal sticks, tiny paper tiles, brushes standing in a jar, and a pile of card ready to fold.

This book is that room. It is called **the Bright Studio**, and it belongs to you for this year.

You do not need to be "good at art" to start. You need curious eyes and hands that want to try.

Four kinds of making live here:

- 1 **Drawing** from things you really meet
- 2 **Mosaic** from small shapes that make a big pattern
- 3 **Painting** from colour, mood and brush marks
- 4 **Structure** from forms you can hold and join

Rufus the fox works carefully with clay and line. Bruno the bear mixes colour slowly and never minds starting again. You will see both of them as you go.

○ DO YOU REMEMBER?

In Years 1 and 2 you made marks, used clay, wove, printed and painted. Keep all of that. This year you look longer, choose materials on purpose, and say what you notice about your own work.

Who you will meet



Rufus the fox

He looks hard before he draws. He likes sharp lines, neat tiles and pots that stand up straight. When something wobbles, he tries one new join.



Bruno the bear

He mixes colour as if he has all morning. He paints big first, then adds small marks. He is good at saying what he likes about a piece of work.



The long table

Everyone works here. Tools have places. Spills get wiped. Finished pieces dry on the side shelf, not on your chair.

You

Age about seven. Ready to look, make, talk and try again. This book is your studio diary as well as your making space.

How to use this book

The same friendly signs come back on every page. Learn them once.

★ SIGNS TO LOOK FOR

- **Open the door** - a question to talk about before you begin
- **Warm up your hands** - a quick game so your hands wake up
- **Look and learn** - something new, with a picture that shows how
- **Your turn** - time to make, draw, cut or build
- **Did you know?** - a true and surprising fact
- **Go a bit further** - an extra idea if you finish
- **Big challenge** - a harder task; take your time
- **How did it go?** - a short look back at your work

When you see a **dashed box**, that space is yours. Draw in it. Stick in it. There are no printed right answers for open making. Say what you think out loud, then choose.



Picture 0.1 Rufus and Bruno read a wooden signpost in the studio. Look at how they stop and check the sign before they choose a way to go, which is what the signs in this book are for.

Getting set up

Your studio kit

You will not need everything at once. Each unit tells you what to fetch.

- pencils (hard and soft), rubber, sharpener
- charcoal stick and a wet wipe for fingers
- black ink or dark washable pen (adult choice)
- scissors (safety), glue stick
- coloured paper or magazine scraps for mosaic
- washable paints, two brushes (round and flat), water pot
- card, tape, string, small recycled boxes
- apron or old shirt
- a sketch pencil case that stays closed when you walk

Room norms

- 1 Look with your eyes before you touch with your hands.
- 2 Share tools. Put each tool back where it lives.
- 3 Keep water pots steady. Wipe spills at once.
- 4 Walk with scissors closed and pointing down.
- 5 Talk about work kindly: what you notice, what you wonder.



Picture 0.2 The kit from the list above, laid out on the long table and seen from directly overhead. Find each one: the pot of pencils, the stick of charcoal, the ink bottle, the scissors and glue, the paint tin with its brushes, the coloured paper with small pieces already cut, and the folded apron. Rufus and Bruno are at the near edge, with their backs to you.

✦ **WARM UP YOUR HANDS**

Shake your hands for five seconds. Press your palms together hard, then open them wide like stars. Do it three times. Now your hands are ready to make.

UNIT 1 · LOOKING TABLE

1

Drawing from encounter

Meet something real. Look longer. Draw what you see.

An encounter is a real meeting: a shell, a face, a leaf, a shoe. We draw from that meeting, not from a guess in our heads.



Picture 1.0 Look at the middle of the long table, where three real things are waiting: a pine cone, a metal spoon and a small plant in a clay pot. Behind them the studio wall is covered with framed drawings of pine branches, all made by looking. Nothing here is a cartoon.

UNIT 1 - LOOKING TABLE

Before you begin

IN THIS UNIT YOU WILL

- ✓ look carefully at a real object before your pencil moves
- ✓ use line, shape, tone and texture because you chose to, not by accident
- ✓ try pencil, charcoal and ink, and say how each one feels in your hand
- ✓ keep a small sketch record of things you meet in your own week

DO YOU REMEMBER?

In Year 1 and Year 2 you made marks, printed, wove and painted. You already know how to hold a pencil softly and how to fill a page.

This year one thing changes. You look first, and you look for longer than feels comfortable. Then you draw.

What to fetch for this unit

- a pencil you like, plus one softer pencil if your class has them
- a rubber and a sharpener
- a charcoal stick, and a damp cloth for your fingers
- black ink or a dark washable pen, if your teacher hands them out
- your sketchbook, or loose paper and a folder to keep it in
- one real object from home or from the school yard, small enough to sit on a table

UNIT 1 - TOPIC 1.1

Looking carefully and drawing what you see

■ Open the door

Hold up one hand. Look at the lines on your palm for ten whole seconds, without saying anything.

What did you notice that you had never named before?

✦ WARM UP YOUR HANDS

Stand up. Shake both hands for five seconds, as if you are shaking off water. Press your palms together hard, then open them wide like two stars. Do that three times.

Now pick up a pencil and draw ten circles in the air above your paper without touching it. Your hand is ready.

► Here is how it works

Observational drawing means drawing from something real that is sitting in front of you, right now.

It is not one action. It is a loop, and the loop is the whole craft.

1 Look

Count something real. Spikes, leaves, holes, corners. Say the number out loud.

2 Draw

Put down the biggest shape first. Do not start with a tiny detail.

3 Look again

Compare your drawing with the object. Change one thing that is wrong.

Most people do step 1 for one second and step 2 for ten minutes. Artists do the opposite.

★ DID YOU KNOW?

A pine cone is a weather machine. Its scales open wide when the air is dry and close up again when the air is damp. A cone you draw on Monday can be a slightly different shape by Friday.

Big shapes first

Before you draw a single edge, find the shape that would hold the whole object if you dropped it in a box.

Make this frame with your fingers. What is the biggest shape inside it?



Picture 1.1 Look at the two spoons on the sheet. On the left is the real spoon, with a curled pattern pressed into the end of its handle. On the right is Bruno's drawing of that same spoon, following the real oval and the real curled end. His paw is still holding the pencil, so the drawing is not finished yet.

■ WATCH ME WORK ONE OUT

Bruno draws the spoon from the long table.

- 1 He finds the long oval of the bowl of the spoon. He draws it lightly.
- 2 He adds the thin handle joining it, in one line.
- 3 He looks again. The handle is too long, so he shortens it.
- 4 He looks a third time and copies the small curled pattern at the end of the handle, which he had not noticed at all until the third look.

He did not invent a spoon from memory. He met this spoon.

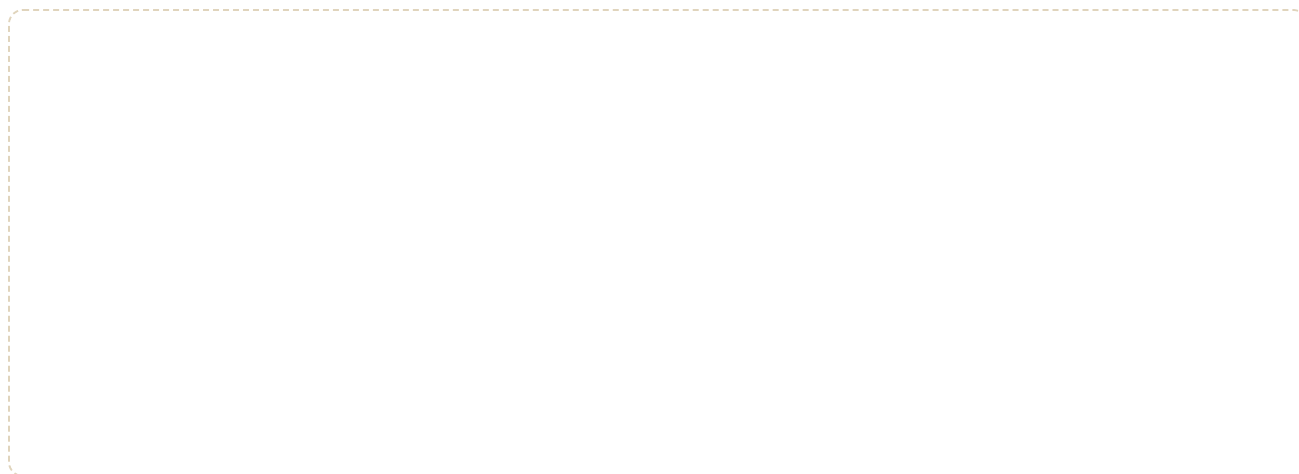
● YOUR TURN

- 1 Choose one real object and put it in front of you. Write its name here.

- 2 Look at it for ten seconds without drawing. Tick everything you can honestly see.

- ☐ a curved edge
- ☐ a straight edge
- ☐ a shadow underneath
- ☐ a shine or a bright patch
- ☐ a small detail you had not noticed at first

- 3 Draw your object in the frame. Biggest shape first, details last.



- 4 Look again for five seconds. Find one thing you got wrong, fix it, and draw a small circle around the fix so you can find it later.

→ GO A BIT FURTHER

Draw the same object a second time, half the size, beside the first one. Small drawings force careful looking, because there is no room to be vague.

▲ BIG CHALLENGE

Turn your object upside down and draw it again without turning your head. Upside down, your brain stops naming the object and starts seeing shapes. Many people find this drawing is their best one.

UNIT 1 - TOPIC 1.2

Line, shape, tone and texture

■ Open the door

Which feels stronger on a page: a thin whispering line, or a thick dark one?

Say which, and say why you think so.

► Here is how it works

Four drawing words work together. Learn one at a time.

WORD	WHAT IT MEANS	TRY THIS
Line	the path a pencil makes	thick, thin, broken, unbroken
Shape	an area that is closed in	circle, triangle, a free shape of your own
Tone	how light or how dark	press softly, press hard, shade in
Texture	how a surface looks as though it would feel	rough dots, smooth glide, scratchy zigzags

● FIND ALL FOUR IN THIS ROOM

Do not draw anything yet. Just look around you and name one real thing for each word. A window frame is a line. A door is a shape.

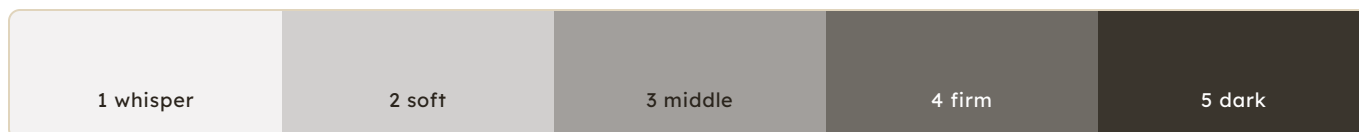
- 1 A **line** I can see from my chair: _____
- 2 A **shape** I can see: _____
- 3 Something with dark **tone**: _____
- 4 Something with rough **texture**: _____



Picture 1.2 Rufus and Bruno lean over one sheet divided into four squares, and every square holds the same bear's head drawn a different way. Look at the top left, built from clean lines. Then the top right, made of flat pale shapes with no outline at all. Then the bottom left, smooth and softly shaded. Then the bottom right, scratched all over with crossing marks.

Tone comes from pressure, not from a different pencil

The same pencil can make five different tones. Your hand decides which.



Build your own ladder in the box below. Match each step, starting as light as you can.

YOUR TONE LADDER: LIGHT TO DARK

✦ DID YOU KNOW?

Your eyes can find a shape with no outline at all. A photograph has no outlines in it anywhere: it is made only of light and dark. That is tone doing all of the work on its own.

● YOUR TURN

- 1 Show **line**, **shape**, **tone** and **texture**, one in each box. The boxes are labelled already.

LINE

SHAPE

TONE

TEXTURE

- 2 Now draw one real thing, a pine cone or a leaf, using at least three of the four words.

- 3 Tick the words you really used, and write the one your drawing needed most.

☐ line☐ shape☐ tone☐ texture

- 4 The word my drawing needed most was _____

- 5 A texture can be lifted straight off a surface. Lay loose paper flat on something rough, hold it still, and rub the side of a soft pencil across it: the surface prints itself. That is a **rubbing**. Take two, from two different surfaces, then bring a small patch of each into these boxes.

PATCH 1: SURFACE

PATCH 2: SURFACE

- 6 Name the two surfaces, then say which patch is rougher. You can tell by looking, without touching either one.

→ GO A BIT FURTHER

Draw the same leaf twice: once with line only, then once with tone only. Show both to someone at your table and ask which one looks more like the real leaf. Their answer is useful even if you disagree with it.

▲ BIG CHALLENGE

Draw your own shoe using **tone only**. No outline anywhere. Press softly for the light leather or fabric, and firmly for the dark sole and the shadow underneath. Stop when someone can tell it is a shoe.

UNIT 1 - TOPIC 1.3

Pencil, charcoal and ink

■ Open the door

You are going out to the school yard on a windy morning to draw one tree. You may take one tool only: a pencil, a charcoal stick or a pen.

Which do you take, and what made you choose it?

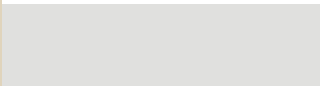
► Here is how it works

Different tools leave different **marks**. None of them is the best one. They are good at different jobs.

TOOL	HOW IT BEHAVES	BEST FOR
Pencil	easy to control, rubs out	detail, and drawings you want to correct
Charcoal	soft, very dark, smudges	big areas of tone, fast and bold work
Ink or pen	brave, cannot be rubbed out	strong edges, and learning to commit

Pencils are marked with letters

Look at the side of a pencil. **H** means hard, and **B** means black. HB sits in the middle. The bigger the number in front of the B, the softer and darker the pencil is.

2H	HB	2B	6B
			
hard and pale	the everyday one	softer, darker	very soft, very dark

★ DID YOU KNOW?

There is no lead in a pencil. The grey core is **graphite**, a soft black form of carbon, mixed with clay and baked. More clay makes a harder, paler pencil.

That is the whole difference between 2H and 6B.

Charcoal is different again: it is simply wood burnt slowly with almost no air.



Picture 1.3 Rufus is pointing at three drawings of one leaf, and Bruno is watching. Look from left to right: the pale pencil leaf with its fine veins, the dark charcoal leaf where the shading does the work, and the ink leaf reduced to one confident outline. The three tools themselves lie along the bottom edge of the table: a pencil, a stick of charcoal and a dip pen.

■ WATCH ME WORK ONE OUT

Rufus draws the same leaf three times, changing only the tool.

- 1 **Pencil.** He draws the veins carefully and keeps the outline light.
- 2 **Charcoal.** He blocks in the dark shape of the leaf with the side of the stick, so the tone carries the whole drawing.
- 3 **Ink.** He draws the outline and the main veins without lifting the pen, and then stops. Nothing can be rubbed out, so nothing is added.

"Same leaf, three voices," he says. "The leaf did not change. I did."

● YOUR TURN

- 1 Make a mark strip. Fill each band with one tool. If you only have a pencil, change how hard you press instead.

LIGHT PENCIL

HARD PENCIL, DARK

CHARCOAL OR A SOFT CRAYON

INK OR PEN

- 2 Now draw one simple object twice, using two different tools.

Tool A: _____ Tool B: _____

- 3 Which tool felt braver, and why?

○ SAFETY AT THE LOOKING TABLE

Charcoal dust stays on your fingers, so wipe them before you touch your face or your paper.

Put the cap back on every pen. Your teacher chooses which ink is safe for your class, and an adult stays nearby while you use it.

→ GO A BIT FURTHER

Draw one object with charcoal, then use a clean finger to smudge one side of it so it looks as though light is falling from the other side.

▲ BIG CHALLENGE

Draw a crumpled piece of paper in ink, with no rubbing out and no second try.

Crumpled paper is all edges and tone. Go slowly, and keep the pen moving.

UNIT 1 - TOPIC 1.4

Sketching your day

■ Open the door

Think about your walk or your ride to school this morning.

What is one small thing you saw that would fit inside a box the size of your hand?

► Here is how it works

A **sketch** is a quick drawing that keeps a moment or an idea. It is not finished art, and it is not meant to be. It is meant to be honest and useful.

Four habits make sketches worth keeping:

- 1 Write the date on the page.
- 2 Draw for three to five minutes, not for thirty.
- 3 Add three words about what you noticed.
- 4 Keep the page even when you dislike it. Sketchbooks are not for showing off.

✦ DID YOU KNOW?

The word **sketch** came into English from the Italian word *schizzo*, which means a rough, quick drawing. Artists have kept sketchbooks for hundreds of years, and most of those pages were never meant for anyone else to see.

◆ ALONG THE WAY

At 10:20 Bruno is sitting on the bench at the tram stop with a small spiral sketchbook open on his knee. The eléctrico is late again. He draws the shelter: the tall post beside him, the roof beams above, and the front of the tram waiting at the corner. He does not draw one cobblestone.

Rufus stands next to the bench with his hands in his apron and waits. "I drew enough to remember the morning," Bruno says. "The cobbles were not the point."



Picture 1.4 Look at Bruno's sketchbook, resting on his knee. The drawing on it is only a few lines: the posts of the shelter and the shape of a building. Then look past Rufus to what Bruno can actually see from that bench, the green and cream tram at the corner under the trees.

A sketchbook page has a shape of its own

The lines at the foot are for three words, not three sentences.

● YOUR TURN

- 1 Sketch one thing from **today** on the page below. Give yourself five minutes, and stop when the time is up. Write the date first, then the three words when you have finished drawing.

SKETCH OF THE DAY	DATE _____

- 2 Tell someone at your table: what did you look at for the longest?

→ GO A BIT FURTHER

Sketch the same corner of the room twice: once standing up and once sitting on the floor. Moving your eyes changes the drawing more than changing your pencil does.

 BIG CHALLENGE

Keep a sketch a day for one week. Small is fine. Five minutes is fine.
On Friday, put the seven pages side by side. You will see your own week,
drawn by your own hand.

UNIT 1 - LOOKING WALL

The Looking Wall

This is the studio show for Unit 1. Your class turns one wall into a wall of real looking.

- 1 Choose **two** drawings from this unit: the one you think is your best, and one that taught you something even though it went wrong.
- 2 On a small card, write the object you drew and the tool you used. Nothing else.
- 3 Put the two drawings side by side on the wall, with the card underneath.
- 4 Walk along the whole wall slowly, once, without talking.
- 5 Now find one drawing by someone else and tell them one true thing you noticed in it. Not "it is nice". Something you actually saw.

★ DID YOU KNOW?

Galleries hang drawings at the eye level of the people who will look at them. In a Year 3 classroom that is lower than an adult would choose, so measure to your own eyes, not to your teacher's.

Saying something true, and saying it kindly

"It is nice" tells the artist nothing. These four openings do. Choose one and finish it out loud, standing in front of the drawing you chose.

- I noticed that you
- The part I looked at longest was
- I want to know how you made
- This made me think of

UNIT 1 - CLOSE

What can you do now?

Tick a box only when it is honestly true. An empty box is useful information, not a failure.

- ☐ I look at a real object before my pencil moves.
- ☐ I start with the biggest shape and add details afterwards.
- ☐ I can name line, shape, tone and texture, and point to each one in my own work.
- ☐ I can make five different tones with one pencil.
- ☐ I can say how pencil, charcoal and ink feel different to use.
- ☐ I can make a five minute sketch of something real from my own day.

● HOW DID IT GO?

My favourite drawing in this unit is the one of

because

The hardest part of looking for a long time was

Next time I draw, the one thing I will do differently is

UNIT 1 - WORDS

Words from the Looking table

encounter

a real meeting with a real thing, which is where every drawing in this unit starts

observational drawing

drawing from something that is in front of you now, not from memory

line

the path your pencil makes as it travels

shape

an area closed in by a line

tone

how light or how dark a part of a drawing is

texture

how a surface looks as though it would feel to touch

graphite

the soft grey carbon inside a pencil, mixed with clay and baked

charcoal

wood burnt slowly with almost no air, used for soft dark drawing

sketch

a quick drawing that keeps a moment or an idea

smudge

to soften a charcoal or pencil mark by rubbing it, on purpose

Say each word aloud once. Then point to the place in this unit where you used it.

UNIT 2 · TILE TABLE

2

Mosaic

Small pieces. Clear gaps. One big picture.

A mosaic is a picture or a pattern built from many small shapes set close together. In Portugal you meet them on walls, on floors and in old railway stations.



Picture 2.0 A wooden tray has been tipped over on the tile table and the pieces have spread out. Look at the four colours waiting there: deep blue, pale blue, dark green, cream and one warm yellow. Nothing is arranged yet, and nothing is stuck down. That is where every mosaic starts.

UNIT 2 - TILE TABLE

Before you begin

IN THIS UNIT YOU WILL

- ✓ look at real mosaics from long ago and from streets near you
- ✓ plan a mosaic on paper before a single piece is glued
- ✓ cut and place shapes so the pattern is yours, not an accident
- ✓ use colour two ways: to agree, and to argue

DO YOU REMEMBER?

At the Looking table you learned to look for a long time before your pencil moved.

The tile table asks for the same patience in a different way. Here you place a piece, step back, and move it again before any glue comes out.

What to fetch for this unit

- coloured paper, or clean magazine pages, or thin card in three to five colours
- safety scissors, and an adult nearby while you cut
- a glue stick
- a pencil and a rubber for the plan
- a sheet of card for the finished piece
- a small tray, box lid or saucer to keep your cut pieces in

UNIT 2 - TOPIC 2.1

Patterns from long ago and nearby

■ Open the door

Where have you seen a wall or a floor made of many small repeated shapes? Say where, and say what the shape was.

✦ DID YOU KNOW?

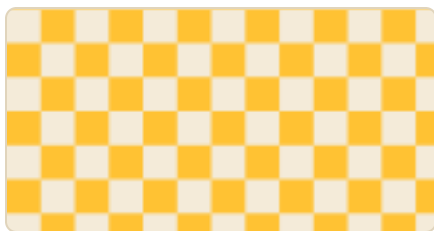
The tiny pieces in a mosaic have a name of their own: **tesserae**. Roman floor makers cut them from stone, and a single floor could use hundreds of thousands of them.

► Here is how it works

People have made mosaics for thousands of years. Roman floor makers set **tesserae**, small cut cubes of stone, into pictures of animals, plants and patterns.

In Portugal the tradition arrives in a different form. **Azulejos** are glazed ceramic tiles, most famously painted in cobalt blue on white, and they cover church walls, staircases, shop fronts and station platforms.

Three things are worth looking for in any mosaic or tile wall:



repeat

one shape coming back again and again



border

a frame that holds the picture in



gap

the thin line between pieces, on purpose

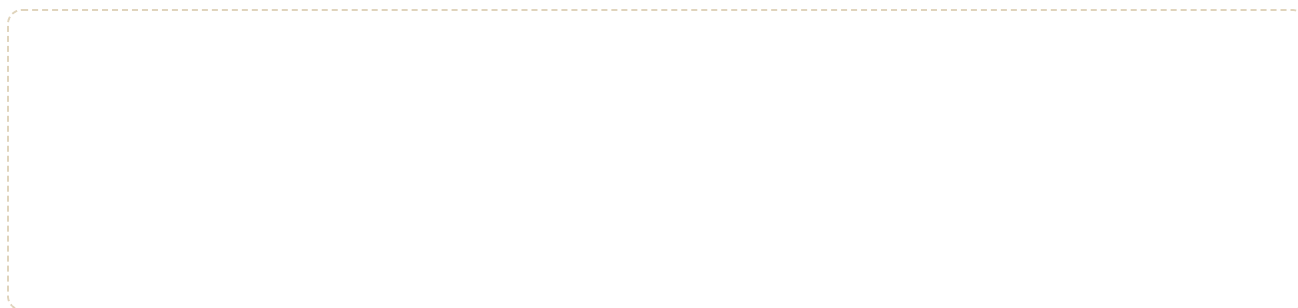
The gap is the part people forget. It is not a mistake and not untidiness: the thin line between pieces is what lets your eye read each piece separately.



Picture 2.1 Rufus and Bruno have stopped in a cobbled street to look up at a wall of azulejos. Look for all three things at once: the flower tile that repeats across the whole wall, the border that frames the panel on the right, and the thin pale lines between every tile. Inside that framed panel is a picture of a building beside water, painted across many tiles at once.

● YOUR TURN

- 1 Look carefully at Picture 2.1, or at a tile wall your teacher shows you. Tick what you can honestly find.
 - ☐ a shape that repeats
 - ☐ a border round the edge
 - ☐ thin gaps between the pieces
 - ☐ a picture made across more than one tile
- 2 Copy **one** small motif from it into the box: a flower, a leaf, a wave, a corner pattern. Use blocks of colour, not smooth shading, because a mosaic has no smooth shading anywhere.



- 3 That pattern feels

because

→ GO A BIT FURTHER

Count how many times your motif would need to repeat to cross a whole wall. Estimate first, then work it out.

UNIT 2 - TOPIC 2.2

Designing a simple mosaic

■ Open the door

Is it quicker to glue first and plan later, or to plan first and glue later? Say what goes wrong if the plan is skipped.



Picture 2.2 Rufus points at the plan while Bruno studies it. Look at the fish drawn on the sheet: its body is already split into angular zones, ready for pieces to be cut to fit. Then look along the right-hand edge, where the colour list is set out as five painted squares, so the choice is made before the cutting starts.

► Here is how it works

Designers **plan on paper** before anything is stuck down, because a glued piece is a decision you cannot take back.

A plan for a simple mosaic needs four things:

- 1 a clear **subject**: a fish, a sun, a letter, a leaf, a flat pattern

- 2 **zones**, so every piece has a home and you are not guessing halfway through
- 3 a short **colour list**, three to five colours and no more
- 4 room for the **gaps**

■ WATCH ME WORK ONE OUT

Rufus designs the fish on that sheet.

- 1 He draws the whole fish lightly, in one outline, no details.
- 2 He splits the body into zones with straight lines, because straight-edged pieces will fit those zones without a fight.
- 3 He picks five colours and paints them along the edge of the sheet as a list: two blues, a green, a cream and a warm yellow.
- 4 He marks where the **biggest** pieces will sit first. Small pieces fill gaps later.

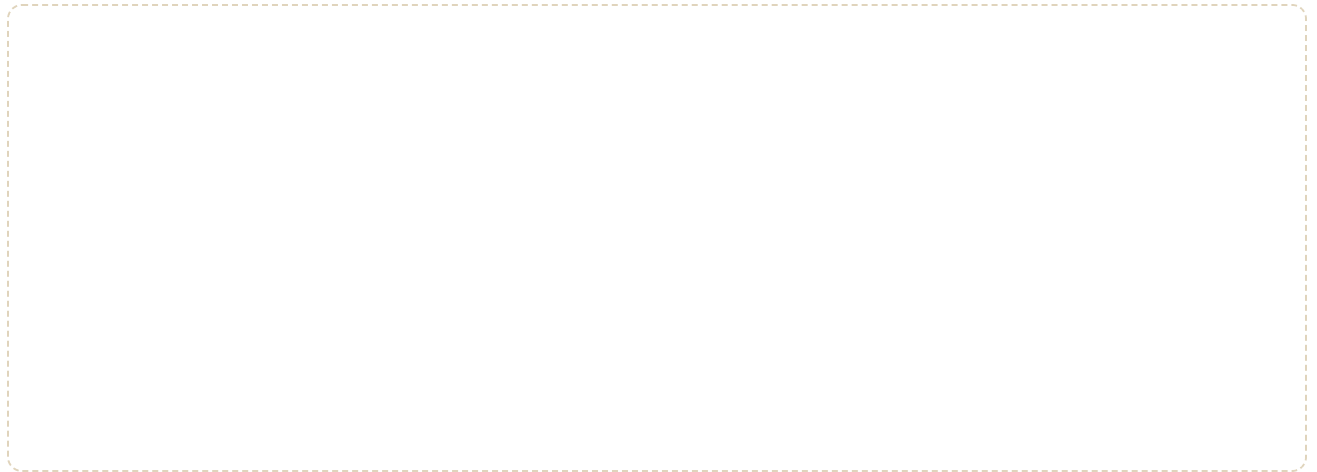
He has still not cut anything. That is the point of a plan.

● YOUR TURN

1 My mosaic subject is

2 My colour list, five at most:

3 Draw your plan in the frame. Keep every shape big enough for your own fingers to cut and place.



4 Mark with an arrow where your biggest piece will go.

→ GO A BIT FURTHER

Add a border to your plan in one colour pair you have not used inside the picture. A border holds a busy pattern together.

UNIT 2 - TOPIC 2.3

Cut, place and pattern

■ Open the door

Why might leaving a small gap between the pieces make a mosaic easier to read, not messier?

► Here is how it works

Four steps, in this order, every time.

1 Cut

Cut a pile of shapes first. Sort them into colours before you start.

2 Place

Lay pieces on the plan with no glue. Move them as often as you like.

3 Look

Step back. Ask someone what they see. Change one thing.

4 Glue

Glue from the middle outwards, a few pieces at a time.



Picture 2.3 Bruno lowers a blue piece onto the plan with one claw while Rufus watches. Look at what has been laid so far: a tree with a bare trunk in the middle of a circle, green pieces gathering into its canopy, a warm orange arc along the bottom and blue beginning at the top. Then look to the left, where the loose pieces sit in bowls sorted by colour. Nothing here is glued yet.

○ SAFETY AT THE TILE TABLE

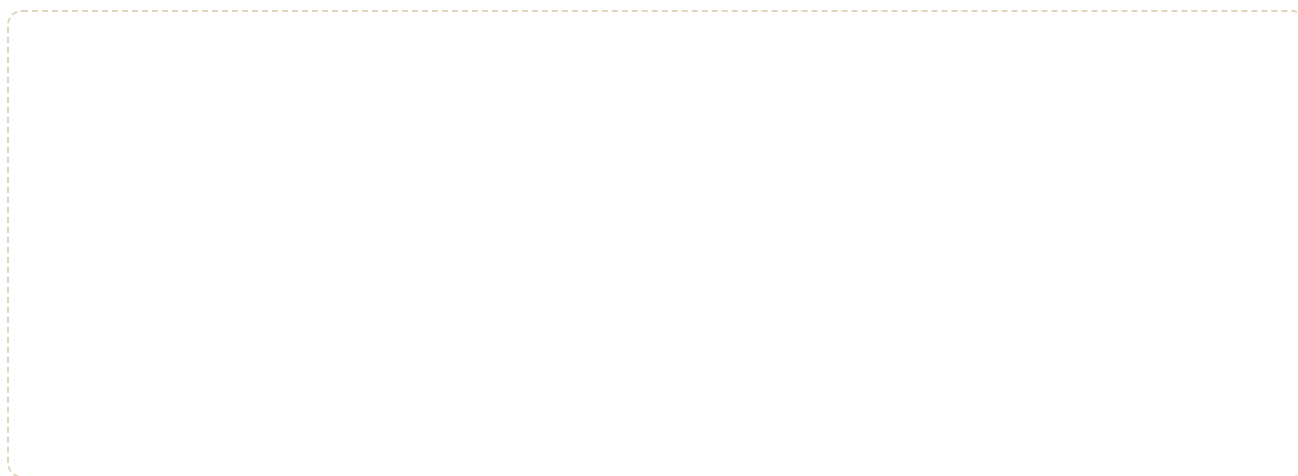
Scissors travel closed and pointing down. Cut with the paper flat on the table, never in the air, and with an adult nearby.

Glue caps click shut. Paper scraps go in the bin, not on the floor, because a floor of paper scraps is a slippery floor.

● YOUR TURN

- 1 Cut at least twelve pieces in two colours. Sort them into two small piles before you place anything.
- 2 Arrange them on your plan. **Do not glue.** Ask a partner what they can see, and write their answer.

-
- 3 Change one thing because of what they said, then glue your final arrangement here.



- 4 The one piece I moved after looking was
-

▲ BIG CHALLENGE

Make a border of your own around the mosaic you have just glued, using only pieces cut from **one** colour, all the same size. A border made of matching pieces is harder than it looks, because every gap shows.

UNIT 2 - TOPIC 2.4

Colour: contrast and harmony

■ Open the door

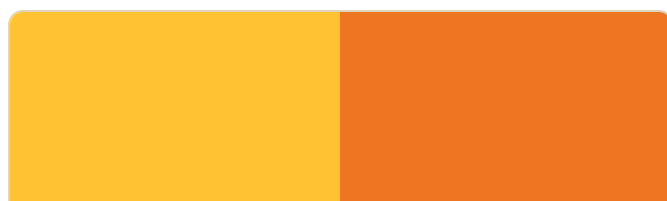
Which pair shouts louder: bright yellow beside deep blue, or pale blue beside pale green? Say which, and say why.



Picture 2.4 The finished sun, and Rufus setting one last yellow piece at its edge. Look at the three rings of decision: the yellows and oranges inside the sun, which agree with each other; the deep blue all around it, which argues with the yellow and pushes it forward; and the plain cream border, which stops the whole disc and hands it a frame.

► Here is how it works

Colours can agree with each other, or argue with each other. A good mosaic uses both, in different places.

**harmony**

colours that sit calmly together, close neighbours, often both warm or both cool

**contrast**

colours that stand strongly apart, warm against cool, light against dark

The usual plan: **harmony inside a shape, contrast at its edge.** That way the shape holds together and still separates from what is behind it.

■ WATCH ME WORK ONE OUT

Bruno builds that sun.

- 1 Centre: yellow and warm orange together, because they agree.
Harmony.
- 2 Around it: one deep blue, chosen because it argues with yellow more than any other colour he has. **Contrast.**
- 3 Outer ring: plain cream, all pieces the same, so the eye stops at the edge instead of wandering off it.

Three decisions, and none of them was about the picture. They were all about colour.

● YOUR TURN

- 1 Fill the two strips: one pair that agrees, one pair that argues.

A PAIR THAT AGREES

A PAIR THAT ARGUES

- 2 Finish or rebuild your mosaic so one area is harmony and one edge is contrast. Write **H** and **C** on your plan to mark them.

- 3 The pair that argues most in my work is _____
and _____

▲ BIG CHALLENGE

Make a small mosaic of your own initials using **three** colours only, with enough contrast that someone can read the letters from an arm's length away.

Squint at it from across the room. If the letters disappear, your contrast is too soft.

UNIT 2 - TILE WALL

The Tile Wall

This is the studio show for Unit 2. Unlike the Looking Wall, this one is built as one piece by the whole class.

- 1 Every pupil brings **one** finished mosaic, glued on card.
- 2 Lay them all on the floor first, not on the wall. Move them around until the colours next to each other look right.
- 3 Choose one colour that appears in nearly every piece. That colour becomes the border of the whole wall.
- 4 Fix the pieces up in the order you agreed on the floor, leaving an even gap between them, because the gap is part of the design here too.
- 5 Stand back together. The class wall is one mosaic now, and each pupil's piece is a tessera in it.

✦ DID YOU KNOW?

The Igreja de São Lourenço in Almancil is lined almost completely with blue and white azulejos, floor to ceiling, telling the story of a saint's life across the walls. A whole room can be one picture.

UNIT 2 - CLOSE

What can you do now?

Tick a box only when it is honestly true.

- ☐ I can find the repeat, the border and the gaps in a real mosaic.
- ☐ I know what tesserae and azulejos are.
- ☐ I plan on paper, with a colour list, before I glue anything.
- ☐ I place my pieces and look again before the glue comes out.
- ☐ I can pick two colours that agree and two that argue, and say why.
- ☐ I leave even gaps on purpose.

● HOW DID IT GO?

The piece of my mosaic I am most pleased with is

The hardest part of working in small pieces was

If I built it again, the one colour I would change is

because

UNIT 2 - WORDS

Words from the Tile table

mosaic

a picture or pattern built from many small pieces set close together

tessera

one small piece in a mosaic; many of them are tesserae

azulejo

a glazed ceramic tile, in Portugal often cobalt blue on white

repeat

a shape that comes back again and again across a pattern

border

a frame around the edge that holds a picture in

gap

the thin line left between pieces on purpose, so each one can be read

zone

an area of a plan that a group of pieces will fill

contrast

two colours that stand strongly apart

harmony

two colours that sit calmly together

motif

the small design that a pattern repeats

Say each word aloud once. Then point to the place in this unit where you used it.

UNIT 3 · PAINT TABLE

3

Painting from encounter

Colour, brush, and a real meeting with what you paint.

We paint what we can look at: a bowl on the table, the view from the window, a friend who will sit still for five minutes. Then we choose the colour and the mark on purpose.



Picture 3.0 The paint table before anyone has started. Look at the saucer on the right: there are only three blobs of paint on it, red, yellow and blue, and every other colour in this unit will be mixed from those three. Then look left, at the bowl of apples, a pear and dark grapes waiting to be painted.

UNIT 3 - PAINT TABLE

Before you begin

IN THIS UNIT YOU WILL

- ✓ paint from something real in front of you, not from a daydream
- ✓ mix your own colours from three, for mood and for distance
- ✓ choose brush marks on purpose and name the ones you used
- ✓ say in clear sentences what your painting is about

DO YOU REMEMBER?

At the Looking table you drew what was really there. At the Tile table you planned before you glued.

The Paint table asks for both at once: look first, decide the colour before the brush touches the paper, and put the big shapes down before any small ones.

What to fetch for this unit

- washable paints, with red, yellow and blue at least
- two brushes: one thicker, one thin
- a water pot, and a second one if you can, so one stays clean
- a saucer, lid or old plate to mix on
- thick paper, and a cloth or paper towel
- an apron, and something real to paint: fruit, a shoe, a plant, a view

UNIT 3 - TOPIC 3.1

Paint what you meet

■ Open the door

Which is easier to paint honestly: a shoe on your desk, or a dragon you have only imagined? Say which, and say what the shoe could teach you that the dragon cannot.



Picture 3.1 Bruno paints the bowl of fruit while Rufus watches. Look from left to right and compare the two: the real bowl on the cloth, with green grapes, dark purple grapes, red apples and a yellow pear, then Bruno's painting of that same bowl. His version is still only blocks of colour. There is not one outline in it, and not one grape has been drawn separately yet.

✦ DID YOU KNOW?

Red, yellow and blue are called the primary colours because you cannot mix them from anything else. Every other colour on your saucer starts with them.

► **Here is how it works**

Painting from encounter means the thing is really there while you paint it. Three classic meetings: a **still life** on a table, a **place** you can see, and a **person** who will sit for a few minutes.

Four steps that keep a painting honest:

1 Look

Longer than feels comfortable. Name three colours you can really see.

2 Block

Big shapes of colour first, thin paint, no outlines, no detail.

3 Deepen

Now the darker parts and the shadows, on top of the blocks.

4 Stop

When the meeting is told. Not when every gap is filled.

■ WATCH ME WORK ONE OUT

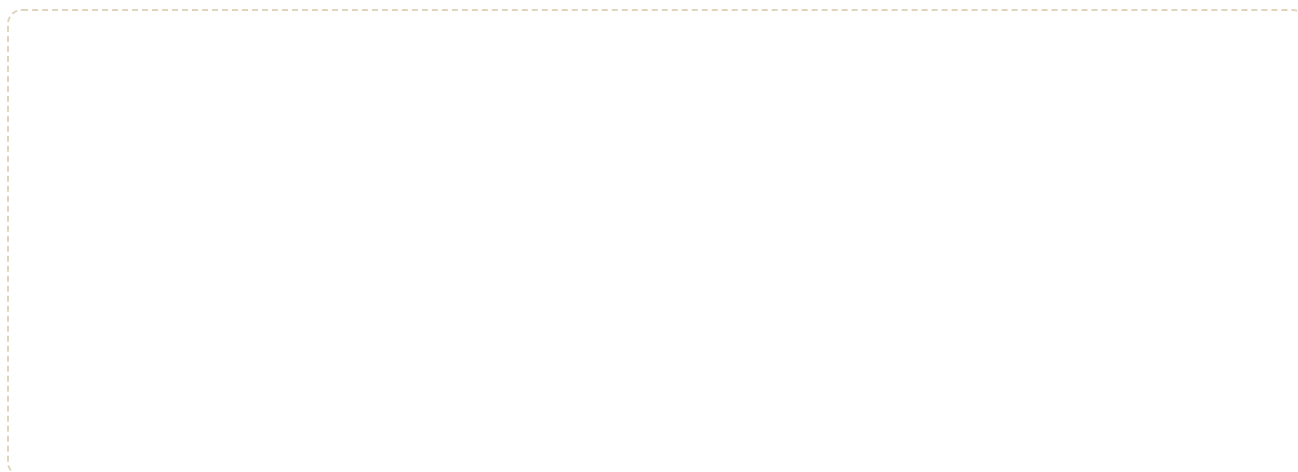
Bruno paints that bowl.

- 1 He mixes a thin grey-brown and lays the whole bowl shape in one move. No handle, no rim, no shine.
- 2 He puts the dark purple mass of the grapes down as **one** shape, not as twenty grapes.
- 3 He blocks the pear in yellow-green and the apples in red, still without edges.
- 4 Only now does he go back for the shine on the apple and the shadow under the bowl.

"If I painted one grape properly first," he says, "the rest would never fit."

● YOUR TURN

- 1 My subject today is (ring one): **still life** / **place** / **person**
- 2 Look at it for a slow count of twenty. Write three colours you can really see in it, not the colours you expect.
- 3 Block the big shapes of colour in the frame. Thin paint, no outlines. Add the darkest part last of all.



- 4 The darkest part of my painting is _____

→ GO A BIT FURTHER

Turn your paper over and paint the same subject again in four minutes flat. A short time forces the big shapes and leaves no room to fuss.

UNIT 3 - TOPIC 3.2

Mix colours for mood and depth

■ Open the door

How is a rainy playground a different colour from a sunny one? Name two colours that change.

► Here is how it works

You have three colours on your saucer. Everything else comes from mixing them.



Mix on the saucer, never in the paint pot. One pot of muddy paint spoils every colour that follows it.

Mood

Mood is the feeling a mix gives. Add water or white and a colour calms down. Add a touch of blue to orange and the fire goes quiet. Add more yellow and it warms up again.

Depth

Depth is making one part feel near and another feel far. Distance does two things to colour: it makes it **paler** and it makes it **cooler**, closer to blue.

FAR: pale, cool, soft edges

MIDDLE: quieter than the front, but still solid

NEAR: strong, warm, clear edges



Picture 3.2 Rufus is painting this one, and Bruno is watching. Look at the three distances on the paper: the dark green wood along the bottom, which is nearest and strongest; the flat blue-grey mountains behind it, paler and cooler; and the almost white sky above. The far mountains are not just smaller. They are quieter in colour.

■ WATCH ME WORK ONE OUT

Rufus paints those two distances.

- 1 Sky first, as a pale wash across the whole top, so everything else can sit in front of it.
- 2 The far mountains next: blue with a lot of water and a little green, edges left soft.
- 3 The near wood last: green mixed strong, hardly any water, and he lets the edge stay crisp.

He did not paint the wood first. It would have made everything behind it look painted on top instead of behind.

✦ DID YOU KNOW?

Air itself changes colour over a distance. Dust and water in it scatter light, so far hills look bluer and paler than near ones even on a clear day. Painters call it **aerial perspective**.

● YOUR TURN

- 1 Mix a **calm** colour and a **loud** colour, and paint a patch of each.

CALM

LOUD

- 2 Paint two hills, one near and one far, using the depth rule: paler and cooler for the far one.

NEAR AND FAR

- 3 My calm colour was made from

▲ BIG CHALLENGE

Paint the same small view twice: once as a bright morning, once as heavy rain. Keep the shapes exactly the same in both, so the only thing that changes is the colour.

Show them side by side and ask someone which is which without telling them.

UNIT 3 - TOPIC 3.3

Brush marks for texture and detail

■ Open the door

Can you make five different marks without changing your brush, only changing how much paint you load and how you hold it?

► Here is how it works

The brush is not one tool. It is five, depending on what you ask of it.

MARK	HOW YOU MAKE IT	GOOD FOR
Wash	plenty of water, one wide move	sky, water, big cloth
Dry brush	almost no paint, drag it	bark, grass, old wood
Dab	tip down and up, again and again	leaves, blossom, crowds
Thin line	the very edge of the brush	twigs, whiskers, wire
Scumble	scrub a broken layer over a dry one	rough walls, stone, fur



Picture 3.3 Bruno points along a sheet of five marks, all made with one brush. Look at them in order from the left: the smooth grey-blue wash, the scratchy ochre patch of dry brush, the little green dabs gathering into a sprig of leaves, the four straight thin lines, and the broken olive scumble at the end.

● YOUR TURN

- 1 Fill the strip with the five marks, in the same order as the table.

WASH

DRY BRUSH

DAB

THIN LINE

SCUMBLE

- 2 Now paint one rough thing, bark or a woolly hat or orange peel, using at least two of the marks.

- 3 The marks I used were

→ GO A BIT FURTHER

Paint the same leaf twice, once with dabs and once with thin lines only.
Which mark suits a leaf better, and why?

▲ BIG CHALLENGE

Paint one small patch of something rough, close up and much bigger than life: a piece of bark, a brick, the sole of a shoe.

Use **three** of the five marks in the same patch. Then hold it up beside the real thing and ask someone which is rougher. If they hesitate, you have won.

UNIT 3 - TOPIC 3.4

Your painting, your response

■ Open the door

If your painting could say one sentence out loud, what would it say?

► Here is how it works

A **personal response** is not "I like it, it is nice." It answers three things:

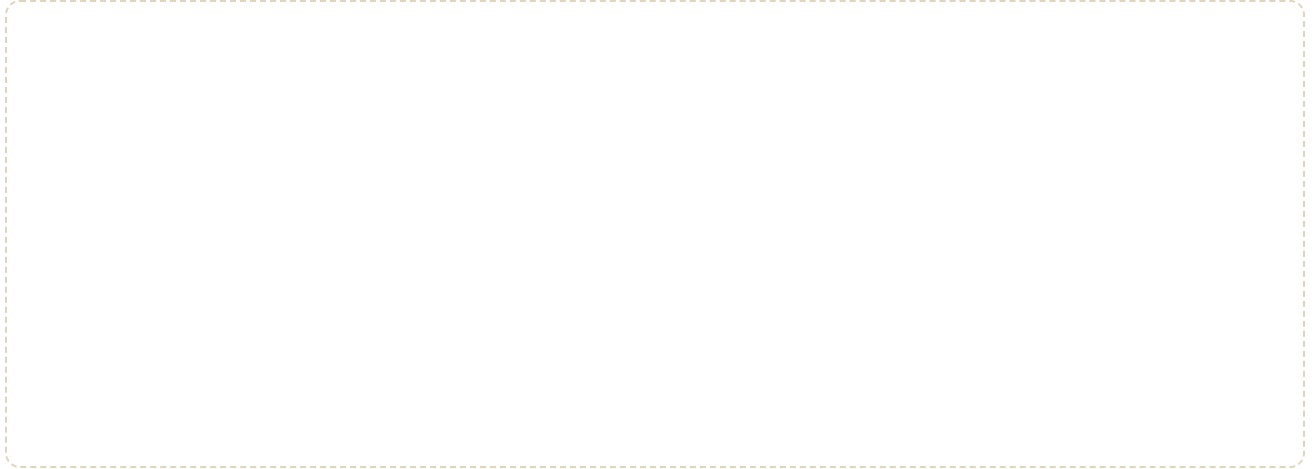
- 1 what you wanted people to notice
- 2 one choice you made on purpose, of colour, mark or where you stopped
- 3 one thing you will try next time

Sentences you can borrow:

- "I wanted people to notice ..."
- "I mixed ... so the mood would feel ..."
- "I used a ... mark here because ..."
- "Next time I will ..."

● YOUR TURN

- 1 Make one finished painting from encounter: still life, place or person. Use what you have learned about blocking, mixing and marks.



- 2 I wanted people to notice

- 3 On purpose I chose

because

- 4 Next time I will

▲ BIG CHALLENGE

Swap paintings with a partner. Without being told anything, write down what you think they wanted you to notice.

Then compare it with what they actually wrote. The gap between the two is the most useful thing either of you will learn today.

UNIT 3 - PAINT SHOW

Two of Everything

This is the studio show for Unit 3, and it is hung in pairs.

- 1 Bring your finished painting, and bring **the real thing** too if you can carry it: the shoe, the plant, the pear, a photograph of the view.
- 2 Set the real thing on a table and pin the painting on the wall directly behind it.
- 3 Walk the room. At every pair, look at the real thing first and the painting second.
- 4 Find one pair where the painting taught you something about the real object that you had not noticed. Tell the painter.
- 5 Nothing here is marked out of ten. The pairing is the point: this whole unit was about meeting something real.

★ DID YOU KNOW?

Painters have set up their own still-life tables for hundreds of years for exactly this reason. A bowl of fruit does not move, does not get bored and does not ask to see the painting halfway through.

UNIT 3 - CLOSE

What can you do now?

Tick a box only when it is honestly true.

- ☐ I paint from something real that is in front of me.
- ☐ I put big shapes of colour down before any detail.
- ☐ I can mix orange, green and purple from red, yellow and blue.
- ☐ I make far things paler and cooler than near things.
- ☐ I can name five brush marks and show each one.
- ☐ I can say what my painting is about in clear sentences.

● HOW DID IT GO?

The painting I would put on the wall is the one of

The colour I was most pleased to have mixed myself was

The hardest part of painting something real was

UNIT 3 - WORDS

Words from the Paint table

still life

objects arranged on purpose so they can be looked at and painted

primary colours

red, yellow and blue, which cannot be mixed from anything else

mix

to make a new colour by putting two together on the saucer

mood

the feeling a colour mix gives to a painting

depth

making one part of a painting feel near and another feel far away

aerial perspective

far things looking paler and bluer because of the air between

wash

thin watery paint laid in one wide move

dry brush

a scratchy broken mark made with almost no paint on the brush

dab

a small mark made with the tip of the brush, down and up

scumble

a broken layer of colour scrubbed over a dry one

Say each word aloud once. Then point to the place in this unit where you used it.

UNIT 4 • BUILD TABLE

4

Exploring structure

Design for a job. Build in three dimensions. Join so it holds.

A structure is anything that holds its shape in space and does a job: a bridge for a toy car, a stand for one tile, a shelter for a small figure in a story.



Picture 4.0 Three forms standing on the build table, and nothing else. Look at them one at a time: a cardboard tube on its end, a grey clay pot pinched into shape by hand, and a small card box with its flap folded shut. Each one holds its shape without help. That is what makes it a form and not a flat sheet.

UNIT 4 - BUILD TABLE

Before you begin

IN THIS UNIT YOU WILL

- ✓ say what job your structure has to do, before you build anything
- ✓ make forms that stand up, in card, in clay and in what you can find
- ✓ join parts so the join can be trusted, and know why it holds
- ✓ test what you built, mend one thing, and test it again

DO YOU REMEMBER?

Three tables so far, and one habit at each: look before you draw, plan before you glue, block before you detail.

The Build table adds the last one. Test before you finish. A structure tells you the truth when you press it gently, and it is the only table where the work can fall over.

What to fetch for this unit

- card, and clean cardboard from a box
- air-dry clay, or plasticine if your class has no clay
- safety scissors, glue stick, sticky tape and string
- a pencil and a rubber for the sketches
- small clean recycled boxes and tubes
- something to test with: a toy car, a small figure, a pebble, one mosaic tile

UNIT 4 - TOPIC 4.1

Design for a purpose

■ Open the door

What is the difference between an object that is pretty and a structure that works?
Can something be both?

✦ DID YOU KNOW?

An arch is one of the oldest strong shapes ever built. It works by passing the weight along the curve and out to the ground on both sides, which is why a stone arch bridge can stand for two thousand years with no glue and no nails in it at all.

► Here is how it works

Purpose is the job the structure has to do. Name it in one sentence, before you touch any card, because everything else is decided by it.

Purposes that work well in Year 3:

- hold one pencil upright without tipping over
- let a toy car roll from high ground to low ground
- keep a small paper figure dry under a spray of water held by an adult
- stand one mosaic tile up so it can be seen from across the room

Then, before sketching, say what it must **not** do: fall, tip, sag, squash, or block the thing it is meant to hold.

Sketch two ideas, not one. One idea is a guess. Two ideas is a choice.



Picture 4.1 Rufus has a paw to his chin, thinking, while Bruno looks at the same sheet. Look at the two pencil ideas below them: on the left an arched footbridge with rails and stone banks, and on the right a shelter with a pitched roof, standing on a raised platform with steps up to it. Both drawings answer the same job. Only one of them will be built.

● YOUR TURN

1 My structure must

2 It must not

3 Sketch two different ideas for that same job. Do not decide yet.

IDEA A

IDEA B

● NOW CHOOSE

1 I choose idea _____
because

2 One thing I will borrow from the idea I did **not** choose:

→ GO A BIT FURTHER

Show both sketches to someone who has not heard your purpose sentence. Ask them what they think the structure is for. If they guess right, your design is speaking clearly.

UNIT 4 - TOPIC 4.2

3D forms in card, clay and more

■ Open the door

Name three things you can hold in your hand that are not flat like a sheet of paper.

► Here is how it works

A **form** has height, width and depth. Paper has only two of those until you fold it.

Five forms worth knowing by name:

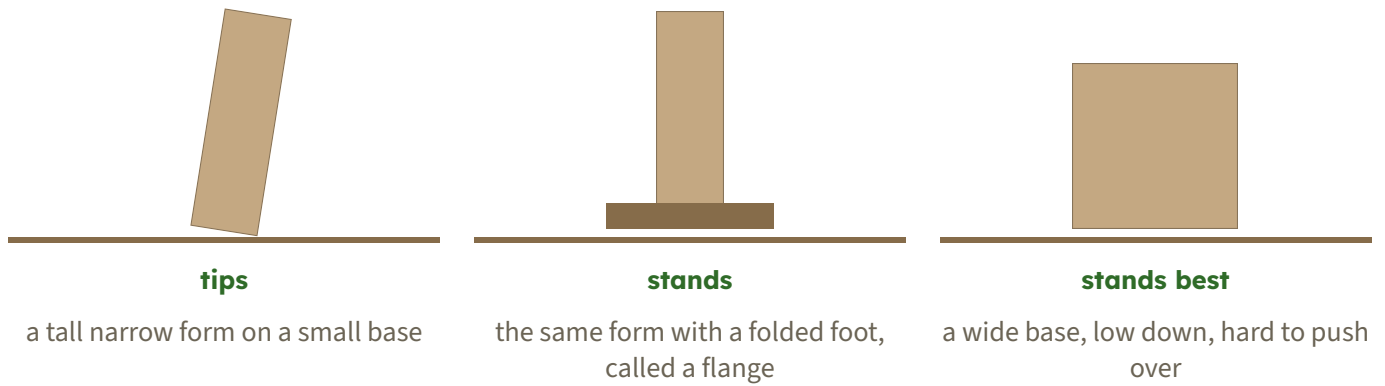
FORM	WHAT IT IS	WHERE YOU HAVE SEEN IT
cube	six square faces	a dice, a small box
cylinder	a tube with round ends	a tin, a kitchen roll
cone	a circle pulled to a point	a party hat, a pine cone
sphere	round the whole way	a ball, an orange
slab	a thick flat wall	a clay tile, a paving stone

And three materials that behave completely differently:

- **card** folds, slots and holds a straight edge, but it hates being wet
- **clay** can be pushed, pinched, coiled and rolled, and it joins to itself
- **found things**, string and sticks and clean boxes, come with shapes already made

A form has to stand

Most structures fail at the bottom, not the top.



Picture 4.2 The same three forms from the unit opener, close up on the bench. Look at the middle one: the clay pot still shows the marks of the fingers that pinched it, which is why it is called a pinch pot. Compare its uneven hand-made wall with the sharp machine-made fold on the card box beside it.

● YOUR TURN

- 1 Make three forms, one from card, one from clay, one from anything you can find. Draw each one after you have made it.

CARD FORM

CLAY FORM

A FORM FROM FOUND THINGS

- 2 Press gently on the top of each one, one finger only. Which held up best?
- 3 Why do you think that one held? Say something about its base.

▲ BIG CHALLENGE

Make a form that stands up on the smallest base you can manage. Measure the base with a ruler and write the width in centimetres.

UNIT 4 - TOPIC 4.3

Joining and building

■ Open the door

For a tall tower, which join would you trust more: a small dab of glue, or a wide tab pushed into a cut slot? Say why.

► Here is how it works

A structure is only as good as its joins. Five worth learning:

JOIN	HOW YOU MAKE IT	BEST FOR
Tab and slot	fold a tab on one part, cut a slot in the other, push it home	card walls and roofs
Flange	fold a small foot outwards along the bottom edge	standing tubes
Tape hinge	a strip of tape across two edges, left slightly loose	doors, lids, flaps
Score and slip	scratch both clay faces, brush on watery clay, press	clay to clay
Lashing	wrap string round and round, tightly, then knot	sticks, rails, tubes

A tab and slot join has one big advantage over glue: you can **test it dry**, take it apart and try again. Glue only gets one chance.



Picture 4.3 Bruno lowers a card roof onto the walls of a small house while Rufus watches. Look along the top edge of the walls, where narrow slots have been cut, and then at the roof panel, which has tabs cut to match them. Look at the front wall too: the window is a hole cut right through, with a cross of card left in place as its frame.

■ WATCH ME WORK ONE OUT

Rufus fits the roof to the walls in that picture.

- 1 He cuts the slots in the wall tops **first**, then makes the tabs to match. Cutting the tabs first means guessing.
- 2 He pushes the roof home with no glue and lets go.
- 3 It stands. He counts three slow seconds, then presses one corner gently.
- 4 Only after it survives the press does any glue come out, and then only inside the joins where it cannot be seen.

○ SAFETY AT THE BUILD TABLE

Safety scissors, tape and glue are the tools for Year 3. Any craft knife is an adult's job, not yours.

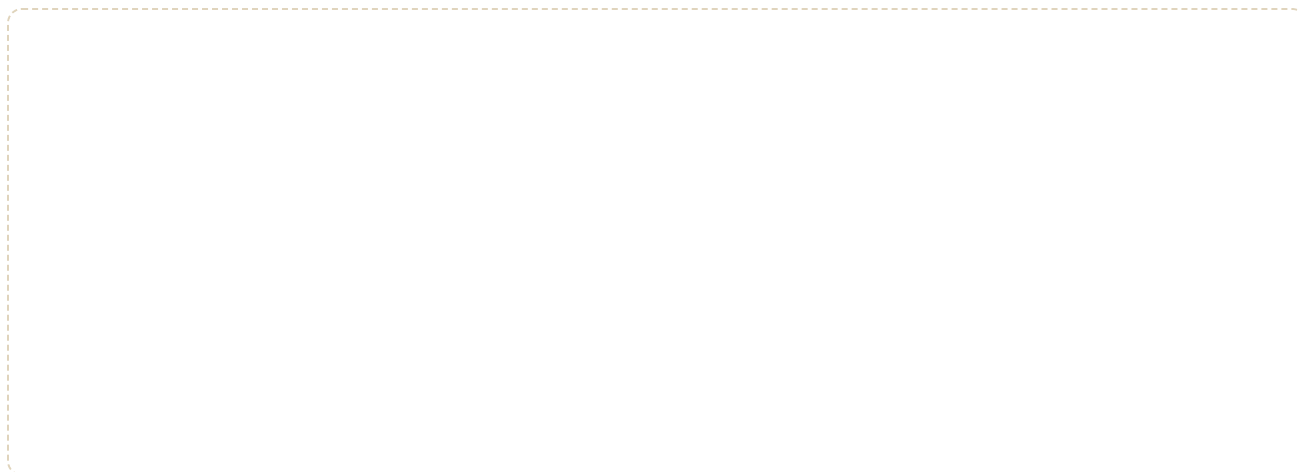
Clay tools stay flat on the table and never near a face. Wash your hands after clay, and before you touch anyone else's work.

● YOUR TURN

- 1 Practise two different joins on scrap card or clay. Name the two you chose.

Join 1: _____ Join 2: _____

- 2 Draw a diagram of the stronger one, big enough to show how the two parts meet. Label the parts.



- 3 It was stronger because

→ GO A BIT FURTHER

Make the same join twice, once carefully and once in a hurry. Pull both apart. The difference you feel in your hands is the whole lesson.

UNIT 4 - TOPIC 4.4

Make a simple structure

■ Open the door

How will you know your structure has worked? Say the test out loud before you build, not after.

► Here is how it works

Build in this order. Structures fail from the bottom, so the bottom goes first.

1 Base

Build and check the base before anything stands on it.

2 Uprights

Add the parts that carry weight, and join them properly.

3 Test

Run the test you named. Gently. Once.

4 Mend

Change one thing, then test again.

One change at a time. If you mend three things at once and it works, you will never know which one fixed it.



Picture 4.4 The bridge from idea A in Picture 4.1, now built and standing. Look for the arch cut from card underneath the deck, the thin sticks lashed with string to make the rails, and the little flight of steps at the right-hand end. Look down as well: the pebbles have been arranged under and around it to stand for the stream it crosses.

● YOUR TURN

1 Build the structure for the purpose you chose in 4.1. Base first.

2 Record what you actually did, not what you meant to do.

Materials:

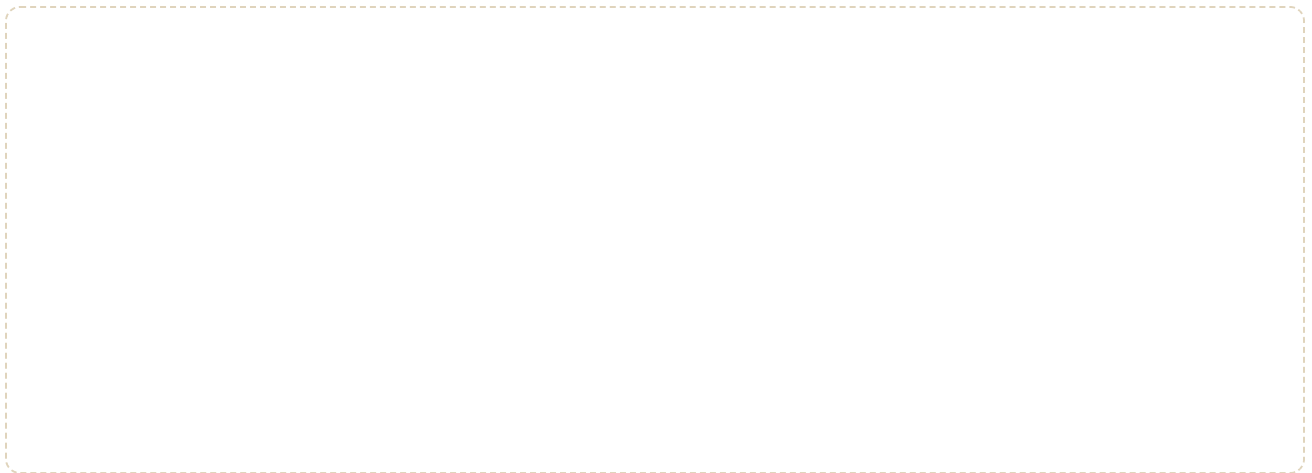
Joins:

3 The test I ran was

and what happened was

4 After the first test I mended

5 Draw your finished structure, from the side, so the joins show.



 BIG CHALLENGE

Build the same purpose again, changing **one** material only. Card instead of clay, or string instead of tape.

Which version was stronger, and was it the material or your second attempt at the joins that made the difference? That question is harder than it looks.

Studio show

Put your year of making on display

This is the show for the whole book, not for one unit. Four tables, four kinds of making, one exhibition.

Choose **four** pieces, one from each unit if you can:

- 1 a drawing made from a real encounter
- 2 a mosaic, or the plan you designed for one
- 3 a painting with your own response sentence beside it
- 4 a structure that passed the test you set it

Exhibition steps

- 1 Write a label for each piece: the title, your name, the materials, and one sentence about what you decided.
- 2 Lay the four pieces out on a table first and choose the order you want people to walk past.
- 3 Set them at your own eye level, not an adult's.
- 4 Practise a talk of thirty seconds about **one** piece. Not all four.
- 5 When people come, ask them "what do you notice?" rather than "do you like it?"

Draft one label here before you write the real ones.

LABEL DRAFT: TITLE, NAME, MATERIALS, ONE SENTENCE

UNIT 4 - CLOSE

What can you do now?

Tick a box only when it is honestly true.

- ☐ I can say the purpose of my structure in one sentence.
- ☐ I sketch two ideas and choose between them.
- ☐ I can name a cube, a cylinder, a cone, a sphere and a slab.
- ☐ I know why a wide base stands better than a narrow one.
- ☐ I can make a tab and slot join and test it dry.
- ☐ I test my structure, mend one thing, and test it again.

● HOW DID THE FOUR UNITS GO?

- ☐ **Drawing:** I can look for a long time and record what is really there.
- ☐ **Mosaic:** I can plan, place and use colour on purpose.
- ☐ **Painting:** I can mix my own colours, choose marks and explain my work.
- ☐ **Structure:** I can design for a job, join, test and mend.

The piece I am proudest of this year is

because

The table I want to go back to is

UNIT 4 - WORDS

Words from the Build table

structure

something built to hold its shape in space and do a job

purpose

the job a structure has to do, said in one sentence

form

a shape with height, width and depth, which you can hold

cylinder

a tube shape with round ends, like a tin

slab

a thick flat wall of clay or card

pinch pot

a clay pot shaped by pressing with the fingers

base

the part a structure stands on, where most failures begin

flange

a small foot folded outwards so an upright will stand

tab and slot

a card join you can test dry and take apart again

lashing

string wrapped tightly round and round to hold parts together

Say each word aloud once. Then point to the place in this unit where you used it.

Studio words

A short list of craft words from this book. Say them aloud. Point to a picture in your work that matches.

WORD	MEANING
observational drawing	drawing from something real in front of you
tone	how light or dark a mark is
texture	how a surface seems to feel
sketch	a quick honest drawing of a moment
mosaic	many small pieces making one picture or pattern
contrast	colours that stand strongly apart
harmony	colours that sit calmly together
still life	objects arranged for looking and painting
wash	thin watery paint in a wide move
dry brush	little paint, scratchy broken mark
depth	near parts strong; far parts quieter
structure	a 3D form built for a purpose
join	how two parts are fixed together
tab and slot	a strong card join
score and slip	a clay join with scratches and watery clay

Answers

Two kinds of task live in this book, and they are answered differently.

Open making has no printed answer. When you are asked to draw your own shoe, choose your own tool or design your own pattern, there is no right version waiting at the back of the book. Your teacher looks at what you decided and why. A page here saying "answers will vary" would be pretending.

Closed questions do have answers, and they are printed below, unit by unit. Check yours after you have tried, not before.

Unit 1 - Drawing from encounter

1.1 Looking carefully and drawing what you see

- The looking loop has three steps: **look**, **draw**, **look again**. The third step is the one most people skip.
- "Big shapes first" means finding the shape that would hold the whole object before drawing any edge or detail.
- Ticking what you can see, and choosing which object to draw, are open: they depend on the object in front of you.

1.2 Line, shape, tone and texture

- The four words are **line**, **shape**, **tone** and **texture**.
- **Line** is the path the pencil makes. **Shape** is an area closed in by a line.

Tone is how light or dark a part is. **Texture** is how a surface looks as though it would feel.

- Five different tones can be made with **one** pencil, by changing how hard you press. You do not need five pencils.
- A photograph contains no outlines at all. It is made only of light and dark, which is tone working on its own.

- Finding all four words in the room has no fixed answer, because it depends on the room. A window frame or a table edge is a line; a door or a book cover is a shape; a shadow under a chair or a dark jumper is tone; brick, bark or a woven basket is texture.
- A **rubbing** is made by laying paper on a surface and rubbing the side of a soft pencil across it, so the surface prints itself. Which of two rubbings is rougher is judged by looking: rougher surfaces print bigger, harder-edged marks.

1.3 Pencil, charcoal and ink

- On a pencil, **H** means hard and **B** means black. **HB** sits between them. The larger the number in front of the B, the softer and darker the pencil.
- So of 2H, HB, 2B and 6B, the palest is **2H** and the darkest is **6B**.
- A pencil contains **no lead**. The grey core is **graphite**, a soft form of carbon, mixed with clay and baked. More clay makes a harder, paler pencil.
- **Charcoal** is wood burnt slowly with almost no air.
- The tool that cannot be rubbed out is **ink**. That is exactly why it teaches you to commit to a line.
- Which tool felt braver is your own answer, and both answers are good ones.

1.4 Sketching your day

- A **sketch** is a quick drawing that keeps a moment or an idea. It is not meant to be finished art.
- The four habits are: **date the page**, **draw for three to five minutes**, **add three words** about what you noticed, and **keep the page** even when you dislike it.
- The word *sketch* came into English from the Italian word *schizzo*.

Unit 2 - Mosaic

2.1 Patterns from long ago and nearby

- The three things to look for in any mosaic are the **repeat**, the **border** and the **gaps**.
- The gaps are there **on purpose**. They are not untidiness: the thin line between pieces is what lets the eye read each piece separately.
- **Tesserae** are the small cut pieces a mosaic is made from. One of them is a tessera.
- **Azulejos** are glazed ceramic tiles, and in Portugal the most famous ones are painted cobalt blue on white.
- Which motif you copy, and how the pattern makes you feel, are your own answers.

2.2 Designing a simple mosaic

- A plan needs four things: a clear **subject**, **zones**, a short **colour list** of three to five colours, and room for the **gaps**.
- Place the **biggest** pieces first. Small pieces fill the spaces that are left.
- Plan before you glue, because a glued piece is a decision you cannot take back.

2.3 Cut, place and pattern

- The order is **cut, place, look, glue** and never any other order.
- Glue **from the middle outwards**, a few pieces at a time.
- Straight-edged pieces fit straight-edged zones without a fight.

2.4 Colour: contrast and harmony

- **Harmony** is two colours that agree, sitting calmly together. **Contrast** is two colours that argue, standing strongly apart.

- The usual plan is **harmony inside a shape, contrast at its edge**.
- In the sun on the tile table: the yellows and oranges inside are harmony, the deep

blue around it is contrast, and the plain cream ring is the border.

- Bright yellow beside deep blue shouts louder than pale blue beside pale green, because it is both warm against cool and light against dark.

Unit 3 - Painting from encounter

3.1 Paint what you meet

- The four steps are **look, block, deepen, stop**.
- Big shapes of colour go down before any detail. Blocking the whole bowl first is what leaves room for everything else to fit.
- Stop when the meeting is told, not when every gap on the paper is filled.

3.2 Mix colours for mood and depth

- The **primary colours** are **red, yellow and blue**. They cannot be mixed from anything else.
- Red and yellow make **orange**. Yellow and blue make **green**. Blue and red make **purple**.
- Mix on the saucer, **never in the paint pot**, because one muddy pot spoils every colour after it.
- Distance makes colour **paler** and **cooler**, closer to blue. Near things are strong, warm and have clear edges.
- Painters call that **aerial perspective**, and it happens because dust and water in the air scatter the light between you and the far hill.

- Paint the sky first, so everything else sits in front of it.
- Which colours you find calm or loud is your own answer.

3.3 Brush marks for texture and detail

- The five marks are **wash**, **dry brush**, **dab**, **thin line** and **scumble**.
- A **wash** is plenty of water in one wide move, good for sky. **Dry brush** is

almost no paint, dragged, good for bark and old wood. A **dab** is the tip down and up, good for leaves. A **thin line** uses the very edge of the brush, good for twigs and whiskers. A **scumble** is a broken layer scrubbed over a dry one, good for rough stone and fur.

- All five can be made with **one** brush. What changes is the paint you load and how

you hold it.

3.4 Your painting, your response

- A personal response answers three things: what you wanted people to notice, one

choice you made on purpose, and one thing you will try next time.

- "I like it, it is nice" is not a response, because it tells the painter nothing.
- The paintings and the sentences are yours, so there is no printed answer.

Unit 4 - Exploring structure

4.1 Design for a purpose

- **Purpose** is the job the structure has to do, said in **one sentence**, before anything is built.
- Sketch **two** ideas, not one: one idea is a guess, two ideas is a choice.
- Saying what it must **not** do (fall, tip, sag, squash) is part of the design.

4.2 3D forms in card, clay and more

- The five forms are the **cube**, the **cylinder**, the **cone**, the **sphere** and the **slab**.
- A **form** has height, width **and** depth. Paper has only two of those until you fold it.
- **Card** folds, slots and holds a straight edge. **Clay** can be pushed, pinched, coiled and joined to itself. **Found things** arrive with their shape already made.
- Most structures fail at the **base**, not the top. A wide, low base stands best; a tall narrow form on a small base tips over.
- A **pinch pot** is a clay pot shaped by pressing with the fingers, which is why the finger marks are still on it.

4.3 Joining and building

- The five joins are **tab and slot**, **flange**, **tape hinge**, **score and slip** and **lashing**.
- A **flange** is a small foot folded outwards so an upright will stand.
- **Score and slip** joins clay to clay: scratch both faces, brush on watery clay, press together.
- A wide tab in a cut slot is stronger than a dab of glue, and it has a second advantage: it can be **tested dry**, taken apart and tried again. Glue gets one chance.
- Cut the **slots first**, then make the tabs to match them. The other way round is guessing.

4.4 Make a simple structure

- Build in this order: **base**, **uprights**, **test**, **mend**.
- Change **one** thing at a time. Mend three things at once and you will never know

which one fixed it.

- An **arch** is strong because it passes the weight along its curve and out to the ground on both sides, which is how a stone arch bridge stands with no glue and no nails.
- What you built, what your test showed and what you mended are your own record.

Checking your own work

Read this part **after** you have tried, never before.

- 1 Check one topic at a time, not the whole unit at once.
- 2 Where your answer is different from the one printed here, do not rub yours out. Write the printed one beside it and put a ring round the part that changed. The two answers together show you what you learned.
- 3 Where the answer above says the response is your own, there is nothing to check and nothing to get wrong. Ask instead: **can somebody else tell what I decided, just by looking?**
- 4 If you disagree with an answer printed here, say so out loud and say why. In art that is a proper argument to have, and you may be right.

Sources and references

Where ideas and facts came from

- School scope and sequence for Year 3 Art & Design (PDF/Input/Student Book/Y3 Art & Design - Student Book.xlsx): used only for unit order and topic scope.
- Studio design traditions in Prime Books Year 1 *The Making Studio* and Year 2 *The Studio by the Wood*: used for house style, not copied sentences.
- General heritage knowledge: Roman mosaic tesserae; Portuguese azulejo tile culture; observational colour change with distance in landscape painting.
- All pupil activities, stories about Rufus and Bruno, and interior illustrations for this title are original to Prime School.

Editorial responsibility

This book is written, illustrated and published by Prime School. The Editorial Board is responsible for the interpretation on these pages.

For teachers

Who this book is for

Year 3 (about age 7). Lower Primary Art & Design. Cambridge Pathway interpretation for Prime School programmes.

How the book is built

Four units match the school scheme:

- 1 Drawing from encounter
- 2 Mosaic
- 3 Painting from encounter
- 4 Exploring structure

Each topic follows a primary studio rhythm: open the door, look and learn, your turn, optional stretch, short reflection. Making space is intentional.

Open creative tasks have no answer key, deliberately. Closed questions do, and they are printed in the Answers section above. Unit 1 is written to that standard; units 2 to 4 are being brought up to it.

Materials and safety

- Washable paints, safety scissors, glue sticks, card, recycled clean boxes, soft charcoal, pencils.
- Adult supervision for cutting and any clay firing discussion. Default clay work is air-dry or adult-fired off-site.
- Risk assessment remains the school's responsibility.

Assessment ideas (light touch)

- Portfolio of four pieces (one per unit) with pupil labels.
- Observation notes: looking time before drawing; care with joins; craft vocabulary used aloud.
- Studio show talk: thirty seconds, one clear intention sentence.

School activities

- Gallery walk in the corridor with labels written by pupils.
- Visit (real or virtual) to a tile museum or local patterned wall, then a response mosaic.
- Outdoor sketching half-hour with clipboards: three five-minute sketches, one chosen to develop.

Design and production notes

- Trim 8.5 × 11 in on every page. White stock. British English. No em-dashes.
- Body type Source Sans 3 at 15 pt, the collection standard for Years 3 and 4.
- Palette sampled from this title's cover (orange spine #EE7623, unit violet

#6B4BA1 at the Looking table).

- Page budget guideline Year 3: about 100-110 pages (DEC-034).